

That Morning Soda Might Be Messing With Your Brain More Than You Think

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Have you ever noticed that **caffeine** makes you feel weird maybe jittery or headachy while your parents just feel more awake? Sophia Zeng, a 16-year-old student in Beijing, wondered the same thing. So she decided to investigate. Her surprising discovery? **Caffeine** might actually change how **genes** work in a young brain, and not in a good way.

Sophia started by digging through old studies on adult mice. She found clues that **caffeine** could switch certain **genes** on or off in the brain. Then she grew rat brain cells in a dish and gave them **caffeine**. Two days later, she looked under a **microscope**. The cells that got more **caffeine** had fewer connections, called **synapses** (the tiny gaps where brain cells talk to each other). It was like the **caffeine** was making the brain's wiring simpler and weaker.

Next, she studied young zebrafish. The fish that got **caffeine** swam more and acted **anxious**, darting into scary parts of their tank. When Sophia checked their brains, four important **genes** for brain growth were less active. That's like dimming the lights on a construction site things don't get built right.

Here's the really interesting part: the gene changes in young fish were different from what she saw in adult rat cells. That suggests **caffeine** hits a young brain harder. Sophia thinks we should be more careful about **caffeine**, especially for kids. "We really don't talk about **caffeine**, even though we drink it," she says. Her work earned her a spot at the huge Regeneron International Science & Engineering Fair, where 1,725 students from around the world shared their discoveries.

Word Treasure

caffeine -- A natural substance found in coffee, tea, and some sodas that makes people feel more awake but can also cause jitters.

jittery -- Feeling shaky or nervous, often like tiny, fast movements in your body.

synapses -- Tiny gaps between brain cells where messages jump from one cell to another.

genes -- Instructions inside every cell that control how living things grow and work.

microscope -- A tool that makes very small things look much bigger so scientists can see them.

anxious -- Feeling worried or scared, often about something that might happen.

Background you need

Many teens drink soda or energy drinks that contain caffeine. Scientists already know caffeine can affect sleep and make people feel jittery.

The Regeneron International Science & Engineering Fair (ISEF) is a huge global science competition where high-school students present their research.

Sophia Zeng's project on caffeine's effect on young brains was one of the entries at this fair, which brings together over a thousand young scientists from around the world.

Break it down

WHO: Sophia Zeng, a 16-year-old student in Beijing.

WHAT: She experimented to see how caffeine changes brain cells and genes in young animals, using rat cells and zebrafish.

WHERE: In her science lab in Beijing, growing rat brain cells in a dish and studying zebrafish in a tank.

WHY: She was curious why caffeine made her feel weird while adults just felt awake, and she wanted to find out if caffeine could be bad for developing brains.

Why it matters

If caffeine can change the way young brains develop, then kids might want to think twice before drinking soda or energy drinks, and it's important for scientists to keep studying everyday substances to keep us healthy.

Test what you remember

1. What did Sophia Zeng first do to start her research on caffeine?

- ☐ (A) She drank lots of soda.
- ☐ (B) She tested caffeine on her friends.
- ☐ (C) She read old studies about adult mice.
- ☐ (D) She built a brain-machine for zebrafish.

2. What happened to the rat brain cells that got more caffeine?

- ☐ (A) They grew bigger and stronger.
- ☐ (B) They had fewer synapses, making their wiring weaker.
- ☐ (C) They turned into fish cells.
- ☐ (D) They started glowing in the dark.

3. How did the young zebrafish act after getting caffeine?

- ☐ (A) They became very sleepy and sank to the bottom.
- ☐ (B) They swam more and went into scary parts of their tank.
- ☐ (C) They turned bright blue.
- ☐ (D) They stopped eating all food.

4. What happened to the four important brain-growth genes in the zebrafish?

- ☐ (A) They became more active than normal.
- ☐ (B) They disappeared completely.
- ☐ (C) They became less active.
- ☐ (D) They turned into fish scales.

5. What does Sophia think we should be more careful about?

- ☐ (A) Eating too much chocolate.
- ☐ (B) Drinking caffeine, especially for kids.
- ☐ (C) Playing video games before bed.
- ☐ (D) Running too fast.

6. What happened to Sophia's project after she finished it?

- ☐ (A) It was hidden in a basement.
- ☐ (B) It earned her a spot at a huge international science fair.
- ☐ (C) It was turned into a TV show.
- ☐ (D) She forgot about it.

Answer key (try the quiz first!): 1: C 2: B 3: B 4: C 5: B 6: B

Questions to think about

1. Positive: Sophia's discovery is exciting because it warns young people about hidden dangers of caffeine and could lead to healthier drinks and better advice for kids.

2. Critical: Others say the study used animal cells and fish, so we don't know for sure if caffeine affects human children in the same way. More research is needed before making big conclusions.

MY THOUGHTS (at least 20 words):